

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2013

CL303.01 Geotechnical Engineering-I

Date: 22.11.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I .

Q - 1 (a) Give the list of some of the important applications in the field of geotechnical engineering. [03]

(b) The following data are obtained in a Compaction test: Determine OMC & MDD. [04]

Water content (w %)	05	10	14	20	25
Bulk density (kN/m^3)	17.7	19.8	21.0	21.8	21.6

Q - 2 (a) Bring out the usefulness of compaction test in the laboratory in soil engineering practice. [04]

(b) Attempt Any Two;

(i) Explain theory of spring analogy for primary consolidation with neat sketches. [05]

(ii) The time to reach 40% consolidation of two-way drained laboratory 10 mm thick saturated clayey soil sample is 35 seconds. Determine the time required for 60% consolidation of the same soil 10 m thick on the top of rocky surface subjected to the same loading conditions as the laboratory sample. [05]

(iii) Give laboratory method to determine coefficient of consolidation. [05]

Q - 3 (a) Give difference between compaction and consolidation. [02]

(b) Attempt Any Two;

(i) A stress at failure on the failure plane in a cohesionless soil mass were; normal stress $= 10\text{kN/m}^2$ and shear stress $= 4\text{kN/m}^2$. Determine the resultant stress on the failure plane, the angle of internal friction and the angle of inclination of the failure plane to the major principle plane. [06]

(ii) Give shear tests based on drainage condition and give brief explanation on the same. [06]

(iii) Three specimens of clay having a small air void content were tested in the shear box. Shear loading was started immediately after the application of normal load and was [06]

completed in 10 minutes. The results obtained were as follows:

Normal stress (kPa)	145	241	337
Shear stress at failure (kPa)	103	117	132

Find the apparent cohesion and angle of shearing resistance of the clay.

SECTION – II

Q - 4 (a) Derive the relation between void ratio, specific gravity, water content and degree of saturation. [04]

(b) Give the general characteristics of the following soils; [03]

(i) Alluvial soils (ii) Aeolian soils (iii) Colluvial soils.

Q - 5 (a) (i) Explain how the flocculated structure changes to dispersed structure, with neat sketches? [02]

(ii) Enlist the uses of particle size distribution curve. [02]

(b) Attempt Any Two;

(i) The sieve analysis test was carried out for 1000 g of soil sample. The result of the sieve analysis test is shown in table below. Plot the particle size distribution curve and find the values of coefficient of curvature and uniformity coefficient. Also identify the type of soil. [05]

IS Sieve (mm)	10	4.75	2.0	1.0	0.600	0.425	0.212	0.150	0.075	Pan
Mass of soil (gm) retained	48	88	149	152	148	88	122	88	71	42

(ii) A soil sample has the liquid limit = 42%, plastic limit = 25%. [05]

% passing 4.75mm sieve = 55%

% passing 0.075mm sieve = 20%.

Classify the soil based on Indian Standard classification system.

(iii) Write short note on Atterberg's limits. How the water content in soil changes different Atterberg's limits? Explain with neat diagram. [05]

Q - 6 (a) The given soil sample is silty-clay having bulk unit weight is 18kN/m^3 and water content is 20%. The specific gravity is 2.70. Calculate void ratio, degree of saturation, saturated unit weight, submerged unit weight and percentage air voids. [05]

(b) Write the validity of Darcy's law. [03]

(c) Attempt Any One;

(i) Write short note on following two terms with neat sketch: [06]

- Quick sand condition
- Piping.

(ii) A horizontal stratified soil deposit consists of three uniform layers of thickness 6m, 4m and 12m respectively. The permeabilities of these layers are 8×10^{-4} cm/s, 7×10^{-4} cm/s and 6×10^{-4} cm/s. Find the effective average permeability of the deposit in the horizontal and vertical direction. [06]